

TWO-WAY COMMUNICATION SYSTEM

25-26J-223

Project Proposal Report (Draft)

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Declaration

I declare that the work presented here is original and that we have not plagiarized in any way, shape, or form any portion of this proposal into another work that has already been submitted for a degree or diploma in any other university or higher educational institution without acknowledgement.

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The supervisor and co-supervisor should certify the proposal report with the following declaration.

The above candidate is carrying out research for the undergraduate dissertation under my supervision.

Signature of the supervisor 

Date 29/8/2025

Signature of the co-supervisor 

Date 28/08/2025

Abstract

This proposal details the development of a “Two-way talking system,” a crucial component within a broader Sinhala Sign Language Communication project. This system is designed to establish a seamless and emotionally-aware communication bridge between hearing and disabled individuals by focusing on bidirectional translation. Specifically, it will accurately convert spoken Sinhala into animated Sign Language, rendered via a virtual cartoon avatar with nuanced facial expressions, in real time. Conversely, it will also facilitate the interpretation of received Sinhala Sign Language back in to spoken Sinhala, enabling true reciprocal communication. Leveraging advancements in speech recognition, natural language processing, and computer animation, this component aims to overcome the limitations of unidirectional communication aids by fostering intuitive, efficient, and emotionally rich interactions, thereby significantly enhancing the quality of life and social integration for the differently disabled community.

Keywords: Sinhala Sign Language, Two-way Communication, Speech Recognition, Virtual Avatar, Real-time Translation

Table of Contents

Declaration	ii
Abstract	iii
TABLE OF FIGURES	v
LIST OF ABBREVIATIONS.....	v
1. INTRODUCTION	1
1.1 Background and Literature Survey.....	2
1.2 Research Gap	3
1.3 Research Problem.....	3
2. OBJECTIVES	4
2.1 Main Objective.....	4
2.2 Sub Objectives	4
2.3 Specific Objectives.....	4
3. METHODOLOGY	6
3.1 Overall System Description	6
3.2 System Architecture	7
3.3 Technical Approach.....	7
3.4 Tools	8
4. PROJECT REQUIREMENTS.....	9
4.1 Functional Requirements	9
4.2 Non-Functional Requirements	9
5. GANTT CHART.....	10
6. BUDGET AND BUDGET JUSTIFICATION	11
REFERENCES.....	12

TABLE OF FIGURES

Figure 3.1 System diagram for the two-way communication system	7
Figure 5.1 Gantt Chart.....	10
Table 6.1 Budget estimation.....	11

LIST OF ABBREVIATIONS

Sinhala Sign Language	SSL
Natural Language Processing	NLP
Text-to-speech	TTL
Deep Learning	DL
Machine Learning	ML
User Interface	UI

1. INTRODUCTION

Communication is the bedrock of human society, fostering understanding, connection, and societal progress. However, for a significant segment of the population-differently disabled and hard of hearing individuals-effective communication with the hearing community remains a profound challenge [1]. This project aims to address this critical barrier by developing a comprehensive, real-time, and emotionally intelligent two-way communication system for Sinhala Sign Language. This innovative solution seeks to bridge existing communication gaps, enhancing social integration, and improving access to information and services for the differently disabled community in Sri Lanka.

This overall project encompasses several critical components working in synergy to achieve seamless communication. These includes:

- Real-time Sign Language Recognition: Accurately interpreting Sinhala Sign Language, including crucial emotional cues.
- Expressive Voice Output: Generating natural Sinhala voice from recognized sign language, compete with emotional nuances.
- Environmental Sound Alerts: Providing crucial notifications about surrounding sounds to enhance the safety and awareness of differently disabled users.
- Two-Way Talking System: Enabling hearing individuals to communicate with deaf individuals by converting spoken Sinhala into visually expressive Sinhala Sign Language via a virtual avatar.

The focus of this proposal specifically delves into the “Two-way talking system” component, which is vital for establishing a truly bidirectional communication flow within the larger framework. This component will allow hearing individuals to speak naturally, with their words being instantly translated and presented in Sinhala Sign Language by an animated avatar, facilitating direct and intuitive interaction with differently disabled users.

1.1 Background and Literature Survey

In Sri Lanka, Sinhala Sign Language serves as the primary language for the differently disabled community [2] [3]. However, a significant portion of the hearing population does not understand SSL, creating a substantial communication divide [2] [3]. This lack of mutual understanding can lead to social isolation, limited opportunities for education and employment, and difficulties in accessing essential public services for differently disabled individuals [4] [5]. The scarcity of qualified SSL interpreters further compounds this issue, with reports indicating a severe shortage, such as only six interpreters available in Sri Lanka as of 2018 [3]. This highlights an urgent need for technological interventions that can facilitate effective and accessible communication.

Historically, efforts to bridge this communication gap have led to the development of various technological aids. These include systems designed for one-way translation, such as converting sign language into text or speech [3] [6], or translating spoken or written text into sign language [7]. While these tools offer some utility, they frequently fall short of enabling the natural, spontaneous, and bidirectional exchanges that characterize human conversation. This limitation is largely due to several factors:

- **Unidirectionality:** Most systems operate in one direction, preventing true conversational flow. For instance, a system translating sign to text doesn't inherently translate text back to sign, or vice versa, creating disjointed interactions.
- **Technical Constraints in Recognition:** Many existing sign language recognition systems face challenges related to varied datasets, technical difficulties in continuous and real-time recognition, and accurately interpreting context-dependent signs [8].
- **Lack of Emotional Nuance:** Sign language is inherently rich in non-manual features, such as facial expressions, head movements, and body posture, which convey crucial grammatical and emotional information [2]. Many current systems often overlook or inadequately represent these non-manual cues, leading to loss of meaning and emotional context in translations. For example, studies on Sinhala Sign Language have emphasized the importance of gestures and expressions in conveying complete messages [2].
- **Limited integration:** Solutions are often disparate, lacking a cohesive platform that integrates multiple communication modalities (e.g., speech recognition,

NLP, sign generation, emotional recognition [2], and environmental sound alerts) into a single, seamless two-way system.

Recent advancements in artificial intelligence, particularly in deep learning, computer vision, and natural language processing, offer unprecedented opportunities to overcome these limitations. Researchers have successfully applied deep learning models to significantly improve the accuracy of both sign language recognition and speech-to-text conversion [9], [10], [11], [12], [13]. The development of sophisticated virtual avatars capable of generating realistic 3D sign language animation, including nuanced facial expressions, is also gaining traction [14], [15]. These technological strides provide a robust foundation upon which a truly comprehensive and user-friendly two-way communication system can be built.

1.2 Research Gap

Despite the advancements in sign language recognition and translation, a significant research gap exists in the development of real time, comprehensive two-way communication systems for Sinhala Sign Language. Most current systems are either unidirectional or lack the nuanced emotional expression vital for complete communication. There is a particular need for a system that not only translates speech to sign but also represents it with an emotionally expressive avatar, allowing for a more natural and empathetic interaction experience. The absence of a robust, accessible, and real-time Sinhala Speech-to-sign language system that accommodates both manual and non – manual aspects (like facial expressions) limits fluid communication and social participation for the Sinhala-speaking differently disabled community [2].

1.3 Research Problem

The core problem this project addresses is the prevailing communication barrier between hearing Sinhala speakers and differently disabled individuals using Sinhala Sign Language, stemming from the lack of an efficient, real-time, and emotionally expressive bidirectional translation mechanism. This leads to social isolation, limited access to information, and reduced participation in daily life for the differently disabled community.

2. OBJECTIVES

2.1 Main Objective

To establish a seamless and emotionally-aware two-way communication bridge between hearing and differently disabled individuals by developing a real-time system that accurately converts spoken Sinhala into animated Sinhala Sign Language via a virtual avatar.

2.2 Sub Objectives

These are the high-level goals set out to attain in this study.

- To develop a robust speech recognition module capable of accurately capturing and processing spoken Sinhala.
- To design and implement an efficient natural language processing pipeline for converting recognized Sinhala speech into a format suitable for sign language generation.
- To create a virtual avatar capable of rendering Sinhala Sign language gestures and facial expressions in real-time.
- To integrate the speech-to-text, NLP, and avatar animation modules to ensure fluid and synchronized two-way communication.

2.3 Specific Objectives

These are the specific aspects of the topic that will be investigated within the main framework of this study.

- To achieve a minimum of 90% accuracy in Sinhala speech-to-text conversion for a defined vocabulary.
- To develop a comprehensive database of Sinhala Sign Language signs and corresponding avatar animations, including emotional cues, that covers common daily conversations.
- To ensure the real-time conversion and animation process has a latency of less than 500 milliseconds to facilitate natural conversion flow.
- To evaluate user satisfaction and the effectiveness of the two-way communication system through usability testing with a diverse group of both hearing and differently disabled participants.

- To optimize the system for efficient performance on mobile devices, considering computational and battery constraints, ensuring a portable and accessible solution.

3. METHODOLOGY

3.1 Overall System Description

The “Two-way talking system” will operate as a full-duplex communication system, facilitating meaningful conversations. It will integrate several key components to enable real-time bidirectional communication. The overall system will comprise:

- **Speech Recognition Module:** This component will be responsible for capturing and transcribing spoken Sinhala into text.
- **Natural Language Processing Module:** The transcribed Sinhala text will be processed to understand its meaning and context, and to prepare it for conversion into sign language glosses or representations.
- **Sign Language Generation Module:** This module will translate the processed text/glosses into specific Sinhala Sign Language commands for the avatar.
- **Avatar Animation Module:** A virtual cartoon avatar will render the generated sign language gestures, along with appropriate facial expressions and body movements, in real-time.
- **User Interface:** A user-friendly mobile application interface will manage inputs, outputs, and overall interaction.

3.2 System Architecture

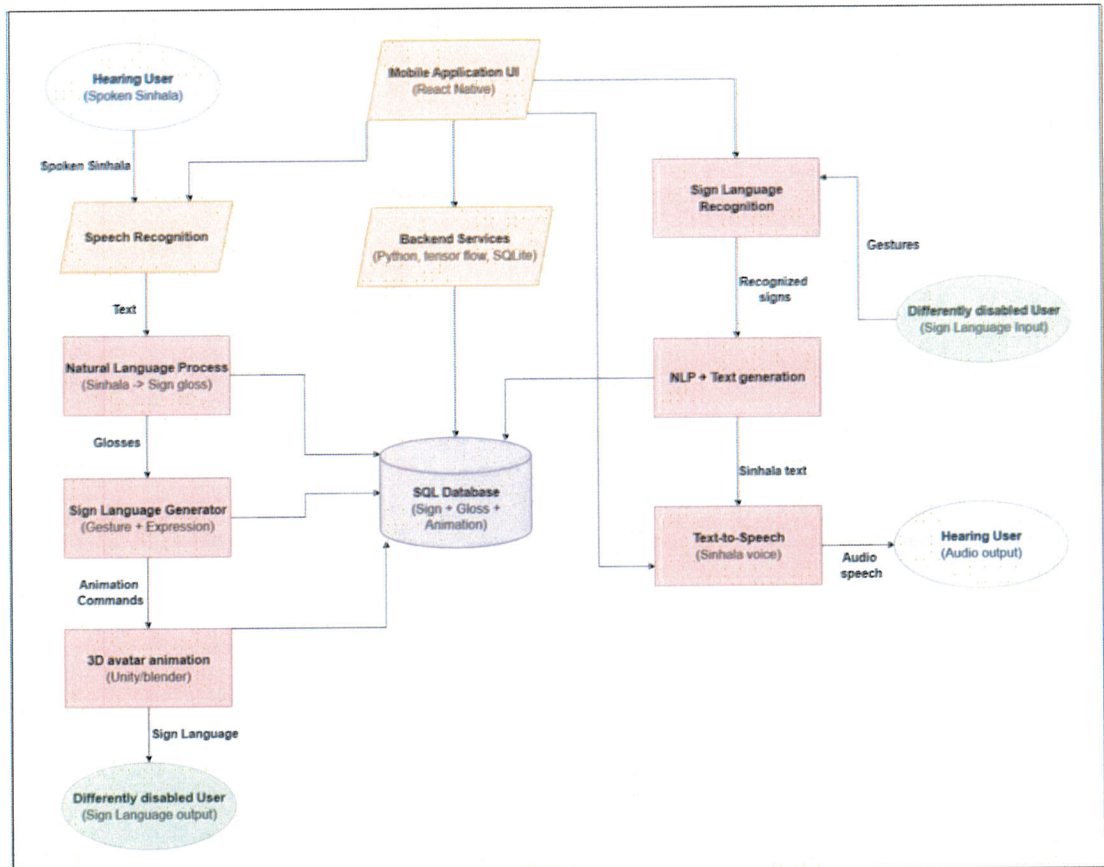


Figure 3-1 System diagram for the two-way communication system

3.3 Technical Approach

The development of this system will involve a multi-faceted technical approach:

Speech Recognition: We will explore deep learning models such as recurrent neural networks or transformer-based architectures for accurate Sinhala speech-to-text conversion. The focus will be on optimizing for real-time performance and handling varied accents and speech patterns.

Natural Language Processing: For Sinhala NLP, techniques will include part-of-speech tagging, dependency parsing, and semantic analysis to accurately convert spoken language intent into sign language concepts. This may involve developing a specific Sinhala gloss-to-text translation model if a direct speech-to-sign generation is not feasible [15].

Sign Language Database and Generation: A comprehensive database of SSL signs will be curated, including static and dynamic signs, and critically, associated facial expressions and body postures to convey emotion [2]. We will leverage techniques for generating realistic 3D avatar animations from these sign representations [14], [15].

Avatar animation: The avatar will be designed to provide real hand movements and facial expressions, such as smiling or looking sad, to convey feelings beyond just words. This will likely involve computer graphics and animation techniques to ensure smooth and natural movements.

Integration and Teal-Time Performance: The integration of these modules will be crucial for achieving real-time, low-latency communication [16]. Optimizations will be made for deployment on mobile devices, considering computational efficiency and battery life [8].

3.4 Tools

- Frontend of the application – React Native
- Backend – python
- Database – SQLite
- Version Controlling – Git
- Model implementation – Machine Learning (ML) and Deep Learning (DL), Text-to-speech (TTS) systems
- Frameworks – TensorFlow, clustering, OpenCV
- 3D avatar – Unity/Blender/Maya

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

- The system shall accurately convert spoken Sinhala phrases into visual Sign language gestures via a virtual avatar.
- The system shall display appropriate facial expressions on the avatar corresponding to the emotional content of the spoken phrase.
- The system shall support real-time processing of spoken input and avatar animation, with minimal noticeable delay.
- The system shall allow for continuous, bidirectional communication between a hearing user speaking Sinhala and a differently disabled user interpreting the avatar's signs.
- The system shall have a user-friendly mobile application interface for easy interaction.
- The system shall include a comprehensive database of Sinhala Sign Language signs.

4.2 Non-Functional Requirements

- Performance:
 - Speech-to-sign conversion latency shall not exceed 500 milliseconds.
 - Speech recognition accuracy for Sinhala shall be at least 90% for common phrases.
 - Avatar animation shall be fluid and natural, running at a minimum of 24 frames per second.
- Usability: The mobile application shall be intuitive and easy for hearing and differently disabled users to navigate.
- Reliability: The system shall be stable and minimize crashes or unexpected behaviors during continuous use.
- Scalability: The system architecture should be designed to allow for future expansion of vocabulary and sign language complexity.
- Portability: The system shall be operable on standard Android and iOS mobile devices.
- Maintainability: The codebase shall be modular and well-documented for future updates and improvements.

5. GANTT CHART

Work	Month						
Project Phase	1	2	3	4	5	6	7
Project Planning and Requirements							
System Design							
Data Collection and Preprocessing							
Model Development							
Application Development							
System Integration and Testing							

Figure 5-1 Gantt Chart

6. BUDGET AND BUDGET JUSTIFICATION

Item	Description	Estimated cost (LKR)
Software tools	Unity/Blender (May need plugins/licenses)	20,000
Data Collection and Processing	Dataset creation for Sinhala speech + SSL gestures (Signer's payments)	40,000
Human Resources	Sign language experts (consultation, validation)	10,000
Testing and Deployment	Usability testing with hearing and differently disabled participants (travel, stipends) Mobile App deployment	6,000
Miscellaneous	Unexpected expenses	5,000

Table 6.1 Budget estimation

Total Estimated Budget: LKR 81, 000

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